[METRIC]
MIL-PRF-62006B(AT)
23 October 1995
SUPERSEDING
MIL-C-62006A(AT)
18 July 1984

PERFORMANCE SPECIFICATION

COOLER, OIL: TRANSMISSION AND ENGINE (METRIC)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers oil cooler assemblies for air cooling engine and transmission oil (see 6.1) and individual cores which makeup the assembly types.

1.2 Classification. The types and styles of oil coolers are as follows (see 6.2):

Type I - Engine and transmission oil cooler end cores.

Class A - Engine only (see drawing 7003292).
Class B - Transmission only (see drawing 7003291).
Class C - Transmission only (see drawing 7985070).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2930

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Type II - Engine and transmission oil cooler side cores.

Class D - Engine and transmission (see drawing 7003293). .
Class E - Engine and transmission (see drawing 7003294).

Type III - Engine and transmission oil cooler assemblies.

Class F - Engine and transmission (see drawing 7003290).
Class G - Engine and transmission (see drawing 8698621).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine Heavy Duty.

(unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 191 11-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

DEPARTMENT OF DEFENSE

7003290 - Cooler Assy, Engine and Transmission Oil.
7003291 - Core, Oil Cooler, End.
7003292 - Core, Oil Cooler, End.
7003293 - Core, Oil Coder, Side.
7003294 - Core, Oil Cooler, Side.
7985070 - Core, Oil Coder.
8698621 - Cooler Assy, Engine and Transmission Oil.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes.

(Copies of ASQC Standards maybe obtained from The American Society for Quality Control, P.O. Box 305,611 E. Wisconsin Ave., Milwaukee, WI 53201 -4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D235 - Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent).

(American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for oil coolers. The materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.7.1).

3.3 Design and construction. All cooler assemblies, coders, and cores shall conform to the manufacturer's design and construction for the applicable item except for outline dimensions; attaching hole dimensions and locations; and inlet/outlet porting holes dimensions and locations which shall conform to the applicable drawing listed in table I (see 4.7.2).

TABLE I. Applicable drawings.

Type of Cooler	Drawing(s)	Component Used On	
		Engine	Transmission
I	7003292	x	
	7003291		x
	7985070		x
II	7003293	x	x
	7003294	x	x
III	7003290	x	x
	8698621	x	x

3.3.1 Flux control. If welding brazing, or soldering is required, all applicable cooler assemblies, coolers and cores shall be free of flux after performance of the fusion operations (see 4.7.4).

3.3.2 Contamination resistance. All cooler assemblies, coolers and cores shall be free of contamination (see 4.7.5).

3.3.3 Interchangeability. All parts of the oil cooler assemblies shall be interchangeable with parts of other manufacturers having the same military part number (see 4.7.2).

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3.3.4 Exterior surface treatment. All exposed exterior surfaces of the cores shall be cleaned, painted, or treated for corrosion resistance as applicable for the materials used (see 4.7.2).

3.4 Performance characteristics.

3.4.1 Leakage. Cooler assemblies, coolers and cores shall not leak when subjected to the following pressures:

3.4.1.1 Pneumatic pressure. Cooler assemblies, coolers or cores shall not leak when subjected to a pneumatic pressure of 689 kilopascals (kPa) [100 pounds per square inch (psi)], when submerged in water at 49 to 60°C (120 to 140°F) (see 4.7.6.1).

3.4.1.2 Hydrostatic pressure. Cooler assemblies shall show no evidence of leakage or permanent distortion when subjected to a hydrostatic dry air or nitrogen pressure of 2758 kPa (400 psi) (see 4.7.6.2).

3.4.1.3 Hydraulic pressure. Cores shall show no evidence of leakage when subjected to 15,000 hydraulic pressure cycles from 0 to 2068 kPa (300 psi) to 0 at the rate of 13 to 20 cycles per minute, at an oil temperature of 77 to 93°C (170 to 200°F) (see 4.7.6.3).

3.4.2 Heat rejection. Heat rejection for the cores and assemblies shall be as specified in table II when using the air and oil properties specified in 4.7.7.

3.5 Identification and marking. Unless otherwise specified in the contractor order (see 6.2), the cooler assemblies or cores shall be marked with the National Stock Number (NSN), the manufacturer's identification code (CAGE), and the part or identifying number (PIN) (see 6.3). Marking shall be permanent and legible (see 4.7.2 and 4.8).

3.6 Workmanship. Workmanship shall be of the quality necessary to produce cooler assemblies and cores free from defects which may affect their serviceability (see 4.7.2 and 4.9).

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TABLE II. Heat rejection for engine and transmission cooler cores and assemblies.

Type	Drawing(s)	Heat rejection-kiloJoules/minute (BTU)	
		Engine	Transmission
I	7003292	2880 (2729)	---
	7003291 7985070	---	6840 (6484)
II	7003293	1430 (1354)	1735 (1643)
	7003294	1335 (1266)	1650 (1563)
III	7003290 8698621	5570 (5280)	10050 (9525)

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on three cooler assemblies or cores when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table III) and the tests of 4.7.6.1 through 4.7.7 (see table IV).

4.2.1 Initial production inspection. Unless otherwise specified (see 6.2), two cooler assemblies or cores from the first 100 cooler assemblies or cores produced shall be selected for initial production inspection. This inspection shall include the examinations of 4.4 (see table III) and the tests of 4.7.6.1 through 4.7.7 (see table IV).

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5 and 4.6.

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TABLE III. Classification of defects.

Category	Defect	Method of examination
Major		
101	Dimensions affecting interchangeability, out of tolerance (see 3.3).	Visual SIE 1/
102	Leakage, oil (external) (see 3.4.1).	Functional
Minor		
201	Identification marking improper (see 3.5).	Visual
202	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	Visual SIE 1/
203	Painted surfaces, finish peeling, blistered or scaling (see 3.3.4).	Visual
204	Workmanship faulty (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE IV. Classification of tests.

Title	Requirement	Test	First article	Quality conformance	
				Acceptance (100%)	Control
Leakage					
Pneumatic pressure	3.4.1.1	4.7.6.1	X	X	
Hydrostatic pressure	3.4.1.2	4.7.6.2	X		
Hydraulic pressure	3.4.1.3	4.7.6.3	X		
Heat rejection	3.4.2	4.7.7	X		X

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ASQC Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table III. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.5 Acceptance test (100 percent). Each cooler assembly or core shall be subjected to the test specified in 4.7.6.1 (see table IV). Failure of any test shall be cause for rejection.

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4.6 Control tests. Control tests shall be conducted on one cooler assembly, cooler or core from each lot of 200 units consecutive] y produced, except that not more than one test shall be performed in a three month period. The cooler assemblies, coolers, or cores shall be subjected to the tests specified in 4.7.7 (see table IV). Failure of any test shall be cause for rejection.

4.7 Method of inspection.

4.7.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3,3.3.3,3.3.4,3.5, and 3.6 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.7.3 Test conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 24 ± 5.5°C (75 ± 10°F), barometric pressure shall be 96.1 + 6.7/-10.1 kPa (28.54 (+2 ,-3) inches mercury), and relative humidity shall be 50 ± 30 percent.

4.7.4 Flux control. To determine conformance to 3.3.1, welded units shall be subject to circulation of water at 82 to 93°C (180 to 200°F) through the core for 10 minutes. Following the water circulation, the chloride content shall not exceed 40 parts per million above existing level for local water.

4.7.5 Contamination resistance. To determine conformance to 3.3.2, cooler assemblies and cores shall be half-filled with mineral spirits per ASTM D235, and capped and shaken in both horizontal and vertical planes. Upon draining through both ports, no particles shall be visible on Whitman #41, or equivalent filter paper.

4.7.6 Leakage tests.

4.7.6.1 Pneumatic pressure test. To determine conformance to 3.4.1.1, each cooler assembly, coder or core shall be submerged in water at a temperature of 49 to 60°C(120 to 140°F). While submerged the internal pressure shall be gradually applied, using dry air or nitrogen, until a pressure of 689 kPa (100 psi) has been reached. The pressure shall be maintained for a period of 1 minute.

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4.7.6.2 Hydrostatic pressure test. To determine conformance to 3.4.1.2 each cooler assembly shall have a pressure of 2758 kPa (400 psi) applied using dry air or nitrogen hydrostatically and maintained for a period of 10 minutes. There shall be no leakage or permanent distortion of the cooler.

4.7.6.3 Hydraulic pressure test. To determine conformance to 3.4.1.3, cores selected at random shall be subjected to 15,000 hydraulic pressure cycles on a test stand capable of maintaining the specified pressure, cycle rate and temperature. Oil used for the test shall be 50 grade oil, conforming to MIL-L-2104. This is a destructive test, so test units shall be indelibly marked “Do not use” and discarded.

4.7.7 Heat rejection test. To determine conformance to 3.4.2, heat rejection tests shall be conducted using oil of the grade indicated entering the cooler assemblies, coolers or cores at the oil flow rates, air inlet and outlet temperatures and air flow rates listed in table V.

4.8 Identification and marking. The cooler assemblies, coolers or cores shall be visually examined to verify conformance to the identification and marking requirements of 3.5.

4.9 Workmanship. Workmanship examinations shall be performed, following phases of the assembly process to verify conformance with 3.6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contractor order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department’s System Command Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Intended use. The coder assemblies or cores covered by this specification are intended for use in military tactical and transport vehicles to cool the engine and transmission oil.

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TABLE V. Air and oil properties used in determining heat rejection.

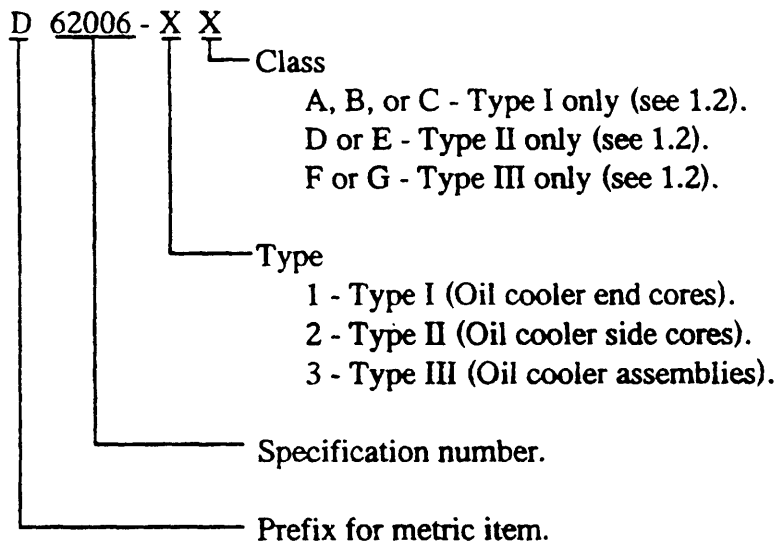
Type	Engine/ Transmission (Drawing)	Air			Oil				
		Flow meter^3/ minute (lbs/min)	Static drop mm H^20 (in. H^20)	Inlet Temperature °C (°F)	Grade (SAE)	Flow liter/ minute (gal/min)	Pressure drop kPa (psi)	Inlet Temperature °C (°F)	Outlet Temperature °C (°F)
I	Engine (7003292)	73 (170)	280 (11)	60 (140)	15W-40	83 (150)	210 (30)	---	107 (300)
	Transmission (7003291 & 7985070)	120 (275)	190 (7.5)	60 (140)	10W	157 (284)	210 (30)	150 (225)	---
II	Engine (7003293)	38 (89)	280 (11)	60 (140)	15W-40	44 (80)	210 (30)	---	107 (300)
	Transmission (7003293)	30 (69)	190 (7.5)	60 (140)	10W	43 (78)	170 (25)	150	---
	Engine (7003294)	36 (84)	280 (11)	60 (140)	15W-40	44 (80)	210 (30)	---	107 (300)
	Transmission (7003294)	28 (65)	190 (7.5)	60 (140)	10W	43 (78)	170 (25)	150 (225)	---
III	Engine (7003290 & 8698621)	387 (900)	280 (11)	60 (140)	15W-40	172 (310)	210 (30)	---	107 (300)
	Transmission (7003290 & 8698621)	387 (900)	190 (7.5)	60 (140)	10W	244 (440)	170 (25)	150 (225)	---

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Type, PIN, and quantity of coolers required (see 1.2 and 6.3).
d. If first article samples shall be furnished (see 3.1).
e. Identification and marking if other than as specified (see 3.5).
f. If first article inspection is not required (see 4.2).
g. If initial production inspection is not required (see 4.2.1).
h. Test conditions if other than as specified (see 4.7.3).
i. Packaging requirements (see 5.1).

6.3 Part or identifying number (PIN). The PINs to be used for coolers acquired to this specification are created as follows:



6.4 Disposition of test assemblies. Assemblies undergoing destructive tests as specified in 4.7.4.3 are not to be used and are to be indelibly marked "DO NOT USE", and purposely scrapped.

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6.5 Subject term (key word) listing.

Contamination
Core
Heat rejection
Pressure

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

The preparing activity must complete blocks 1, 2. 3, and 8 In block 1, both the document number and revisio letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification o requirements on current contracts. Comments submitted on this form do not constitute or imply authorization t waive any portion of the referenced document(s) or to amend contractual requirements. FSC 2930

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62006B(AT)	2. DOCUMENT DATE (YYMMDD) 951023

3. DOCUMENT TITLE
Cooler, Oil: Transmission and Engine (Metric)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(810) 574-8745	786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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